coursera | UNIVERSITY OF MICHIGAN

Applied Information Extraction in Python

Today's goal • 2/3 items

Module 1
Information Extraction

Module 2
Named Entity Recognition (NER)

Named Entities and Named Entity Recognition

- What is Named Entity Recognition (NER)? Video • 5 min
- Knowledge Check: Named Entities and Named Entity Recognition Practice Assignment • Grade: 100%

Setting up NER as a Machine Learning Task

- BIO Encoding for Named Entity Labels Reading • 10 min
- BILOU Encoding for Named Entity Labels Reading • 10 min
- Machine Learning Fundamentals: How Machines Learn to Label Named Entities Reading • 10 min
- NER as a Sequence Classification Task Video • 8 min
- Knowledge Check: Setting up NER as a Machine Learning Task Practice Assignment • 10 min

Hidden Markov Models (HMMs)

- Markov Chain and Hidden Markov Models Reading • 10 min
- Fundamentals of Markov Chain Models Video • 7 min
- Hidden Markov Models (HMMs) Video • 11 min
- Training Hidden Markov Models: How HMMs Learn to Assign Labels Reading • 10 min
- The Math Behind HMMs: How Probabilities Power Sequence Labeling Reading • 10 min
- Conditional Random Fields (CRFs) Video • 6 min
- Knowledge Check: Hidden Markov Models (HMMs) Practice Assignment • 15 min

How to Extract Information Using CRFs in Python?

- Demo: CRF Model Training Video • 8 min
- Jupyter Notebook Practice on Training CRFs Lab • 1h

Module Wrap-Up & Assignment

- Assignment 2 Introduction: Implementing a CRF Model Video • 4 min
- Build an Information Extraction Pipeline for CRF Based Extraction Programming Assignment • 3h
- Module 2 Assignment Graded Assignment • 30 min

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

BIO Encoding for Named Entity Labels

0:08 / 1:31 2x

Why Does Label Encoding Matter in Named Entity Recognition?

In Named Entity Recognition (NER), the goal is to assign a label to each word (or token) in a sentence that reflects whether it is part of a meaningful entity, like a person name, location, organization, or something domain-specific like a drug or disease.

To do this effectively, especially in machine learning-based models like Conditional Random Fields (CRFs), we need a consistent way to represent which tokens belong to which entities, and how.

That's where encoding schemes like **BIO** and **BILOU** come in. These systems give structure to your labels, allowing models to distinguish the beginning, inside, and outside of multi-word entities.

What Is BIO Encoding?

BIO stands for:

- **B**eginning of an entity,
- **I**nside of an entity (after the first word)
- **O**utside any entity

Example 1:

"The University of Michigan is located in Ann Arbor."

Suppose:

- "University of Michigan" is an organization (ORG), and
- "Ann Arbor" is a location (LOC).

Using **BIO**:

- "The" → **O**
- "University" → **B**-ORG
- "of" → **I**-ORG
- "Michigan" → **I**-ORG
- "is" → **O**
- "located" → **O**
- "in" → **O**
- "Ann" → **B**-LOC
- "Arbor" → **I**-LOC

This scheme marks the start of each entity and continues with "**I**-" for inside tokens.

Example 2:

"Google is hiring."

Here, "Google" is a single-word organization (ORG).

Using **BIO**:

- "Google" → **B**-ORG
- "is" → **O**
- "hiring" → **O**

BIO doesn't indicate whether "Google" is a one-word entity or just the beginning of a multi-word one.

Next, we will explain the **BILOU** encoding.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →